Ridge Regression, Gradient Descent, And Perceptron

CSAI 253 - Spring 2024

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Revision of Standard Linear Regression Model

3

pending recreation

4

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Problems of Linear Regression

(With Linearly Separable Data)

Problems of Linear Regression

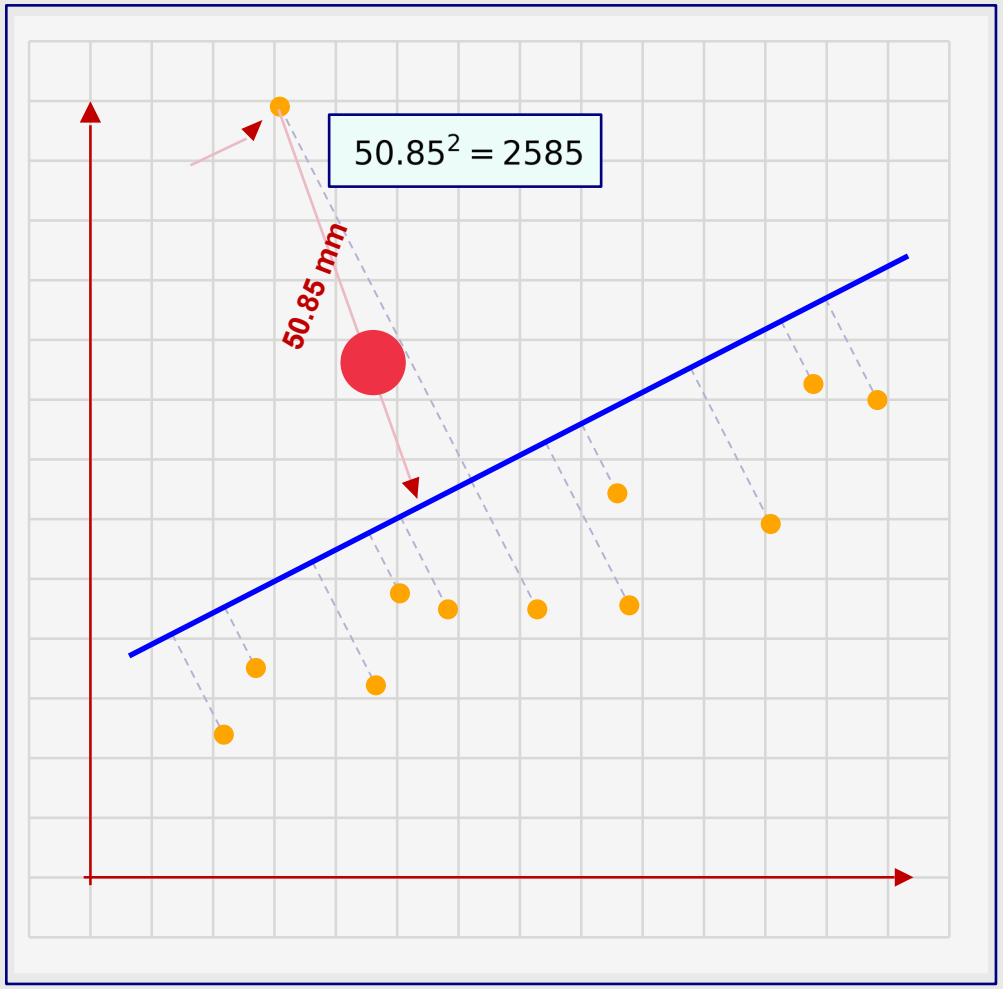
Assume the Dataset is linearly separable (suitable to apply Linear regression model)

Linear regression model has FOUR problems:

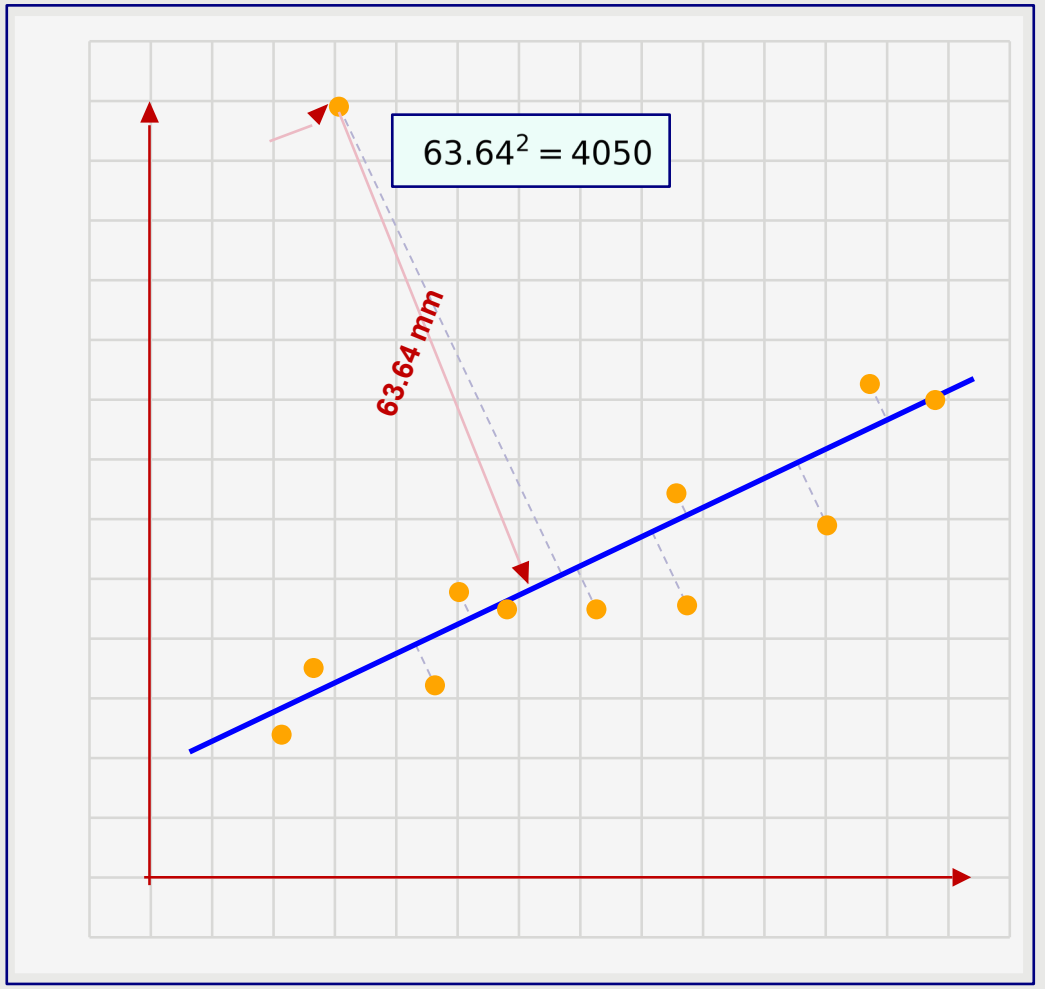
- **Overfitting Problem**
- **Matrix Inversion Problem [1]** ***(Singular Matrix)***
- **Matrix Inversion Problem [2]** ***(Large Matrix dimension)***
- **Different Hyperplanes can solve the problems** ***(Selection of Best Hyperplane)***

[1] Ridge Regularizer to Minimize Overfitting

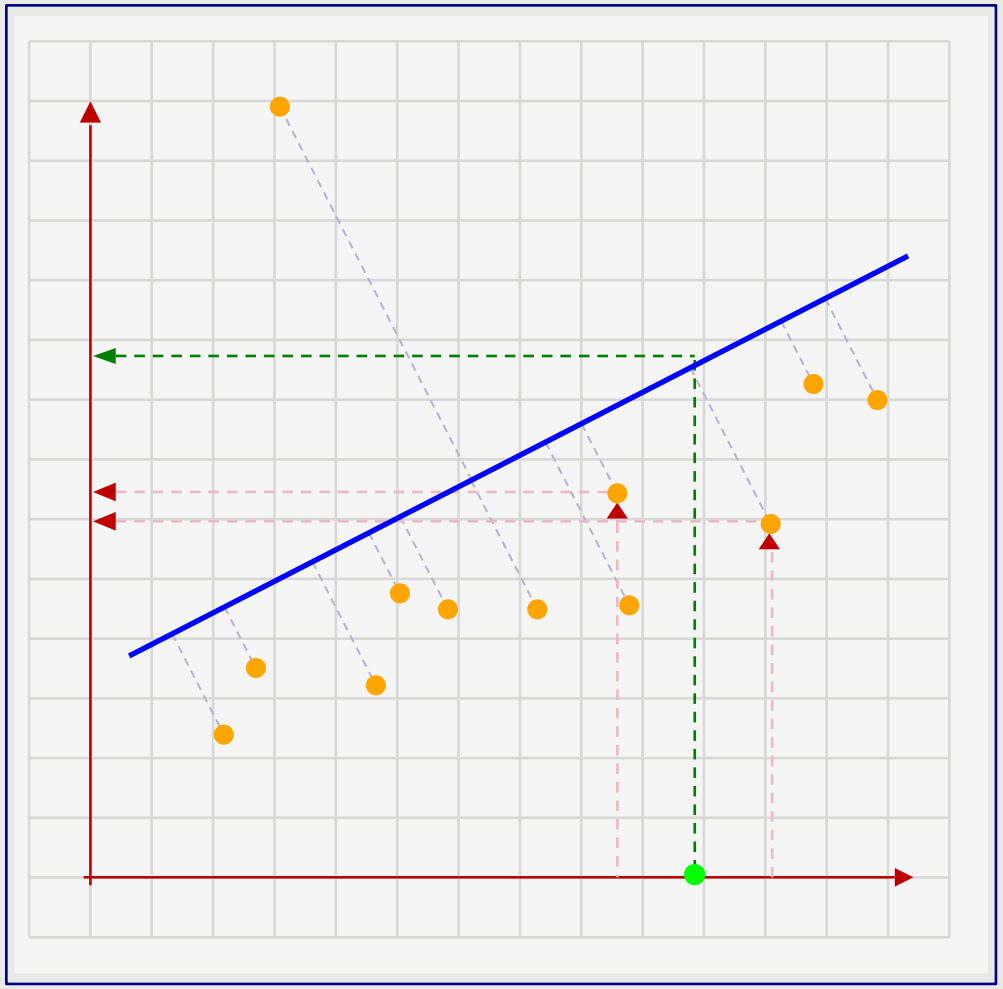
(Also Known as L2 regularizer)



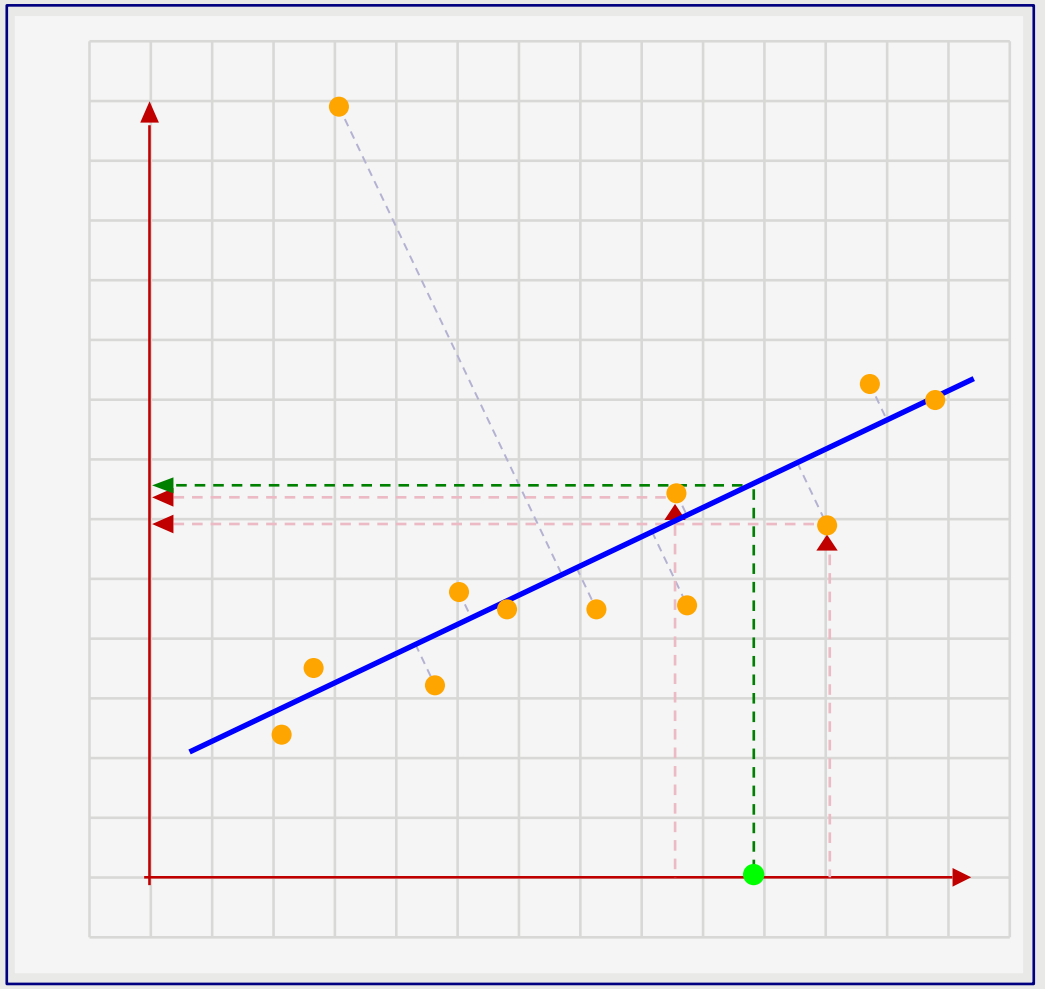
Overfit (try to minimize the residuals ONLY)
Fails to detect the Pattern



Generalized (Residuals are not minimized)
Successes to detect the Pattern



Overfit (try to minimize the residuals ONLY)
Fails to detect the Pattern



Generalized (Residuals are not minimized)
Successes to detect the Pattern

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